P07 | Contribution, evidence and limits

For David Celeita and Masoud Barati | Coauthor-review draft | 02 Nov 2026, 17:26 Colombia

Question and actual contribution

How can eight PMUs distinguish the physical cause of a network disturbance? The paper connects network-constrained reconstruction, physical response signatures and Bayesian history attribution, with engineered-feature ExtraTrees as a retained alternative. The implemented causal bank predicts each candidate intervention from estimated past states, integrates load severity and retains recovery. Active and cause masses expose its decisions. These are approximate conditional calculations, not a new Bayes theorem or a globally optimal filter.

Earlier evidence is part of the paper

Separate component tests recover hidden voltages and measure feature gains; the mixed-source sensitivity test improves local response fit. Historical transfer yields 0/96 joint source decisions versus 96/96 for an earlier physical estimator, with its confusion matrices included. The archived batch support study retains an evidence-normalization caveat. None isolates a tensor/Bayes contribution to final causal accuracy; the causal ExtraTrees tests are distinct from the original 30-s submission.

Sealed comparison: 171 physical trajectories, 342 observed records

Frozen continuous method	Timely / 648	Extras	Outside-bank timely / 216	Outside-bank extras
Physical bank (5 models)	644/648	72	212/216	68
Physical bank (nominal)	643/648	92	211/216	86
ExtraTrees reference	330/648	29	110/216	10
ExtraTrees short	352/648	16	117/216	6

Other outcomes that change the interpretation

Continuous: the physical bank attributes 322/324 loads and 322/324 lines before OFF; the ExtraTrees reference attributes 15/324 loads and 315/324 lines; the short-window variant gets 30/324 loads and 322/324 lines. All timely matched decisions have correct family/source: the main gap is timely coverage, not conditional classification accuracy. Persistent line: both physical banks and both ExtraTrees adapters make 162/162 correct first decisions; Chen2016 makes 74/162, with 66 pre-change stops. After first detection, physical banks produce 0 extras versus 164/179 for the two trees (post hoc persistence audit). Chen stops on all 72/72 normal records; physical banks and trees alarm on 0/72, with different exposure. Restoration audit (posthoc): early associated restorations were 0/644 for the physical bank versus 52/330 for ExtraTrees (26/352 short); missing restoration values were 4/648 versus 318/648 (296/648 short).

Interpretation and limits

The sealed study supports timely attribution on the declared nine-load/nine-line simulator catalog, especially loads, while exposing an alarm burden under network mismatch. Replayed conditions and noise share parents; 648 is not 648 independent events. Original tree validation shares two normal groups with fitting; the later grouped 70/30 study is a different model. No official hackathon-score improvement, source-excluded transfer, all-family diagnosis, calibrated probabilities, field operation or real-time throughput is established. Warner trajectory search and IMM/multiple-history methods are acknowledged; no matched numerical advantage over them is claimed.

For David and Masoud

Coauthor-review draft. A controlled 24-record factorial study is running with the same five-model bank and publisher: marginal versus profile amplitude scores, and physical-state carry versus reset at inferred closure. No effect is claimed before completion; this posthoc study is not independent confirmation. Pending: complete ablation, final integrated-PDF review and author decisions.